Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_03f45ad437df4908bd92610d827648d9
Date	2026-11-01 17:56:51
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	95%	9.0 mm long

2. Root cause

Cause: Hardness Variation in Incoming SPCC Material Lot

Reason: The detected critical crack during bending of SPCC steel is directly explained by incoming material lot hardness variation, which reduces ductility and causes susceptibility to cracking under forming operations [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius, surface defects from work hardening inconsistency

Confidence: 95%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield strength, tensile strength, and hardness
2. Conduct incoming material testing on the affected SPCC steel lot

Preventive actions (stop it recurring)

1. Enforce strict incoming material inspection and hardness verification controls for cold-rolled SPCC sheet metal lots

Precautions

- Ensure incoming material lots are verified against mill test certificates before releasing to forming operations

How to verify the fix

Confirm incoming lot hardness meets specification and sample bend testing shows zero cracking

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Work Hardening Inconsistency in Cold-Rolled Sheet Metal	95%	Inconsistent work hardening characteristics in SPCC cold-rolled sheet metal lead to localized brittleness and cracking at the bend radius during bending operations [02cf0928-6739-4f7e-ae1b-f6740283f00f]. Note: retrieved evidence is limited to this material inconsistency analysis.

3	Out-of-Specification Mechanical Properties (Yield/Tensile Strength Variation)	95%	Mechanical property variation (yield and tensile strength) relative to material specification causes unpredictable forming behavior and cracking during bending [02cf0928-6739-4f7e-ae1b-f6740283f00f]. Note: evidence is limited to document 02cf0928-6739-4f7e-ae1b-f6740283f00f.

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.